late everything else. This approach works passably for global consumer applications. In India, it breaks down. Translation treats language as text. India speaks in context.

Consider a health query phrased by an ASHA worker in colloquial Hindi: "*Bachcha doodh pee raha hai, par wazan kyun nahin badh raha*?" A literal translation—"The child is drinking milk, but why is the weight not increasing?"—captures the words, but not the intent. What she is really asking is whether this pattern signals a feeding issue, an infection, a maternal nutrition problem, or the need for escalation. A generic translated response may return growth charts or calorie norms—technically correct, but practically useless.

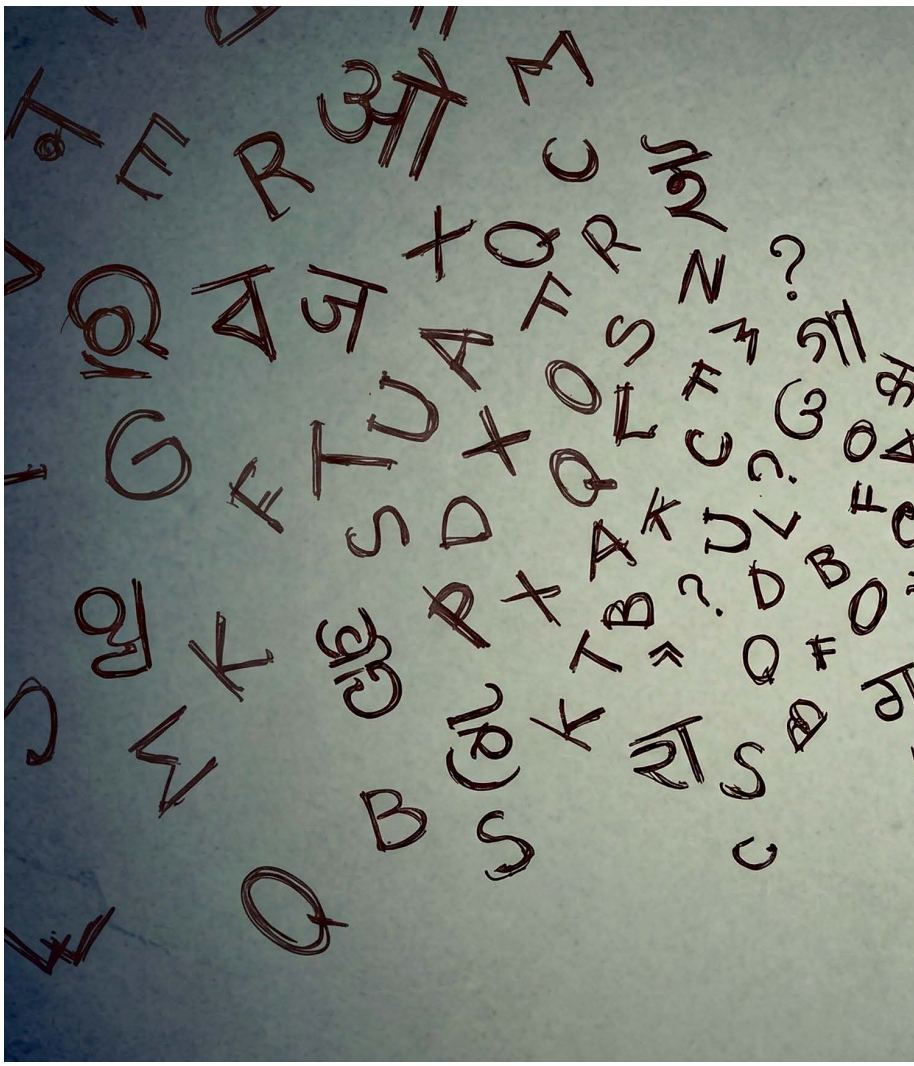
In the field, meaning lives between the words: in tone, in what is left unsaid, in shared assumptions about age, diet and risk. Translation flattens uncertainty into text and strips away the conversational cues that guide real decision-making. The response sounds fluent. The advice feels confident. And yet, the system has misunderstood the question that mattered. In high-stakes domains such as healthcare, governance or education, this is not a technical limitation. It is a safety risk.

Building Foundations

This realisation has begun to reshape how language AI is imagined in India—not as a feature to be bolted on, but as shared infrastructure that must work across languages, regions and contexts. This shift is now being formalised through national efforts such as the IndiaAI Mission, which places language capability, compute, and public-interest deployment at the centre of India's AI strategy. But Indian-language AI did not emerge from a vacuum.

Over the past few years, AI4Bharat has played a pivotal role in changing how Indian-language AI is built and deployed. Its most important contribution has been to demonstrate—through data and models, not rhetoric—that Indian languages can be first-class citizens in modern AI systems. By releasing large-scale, open datasets across text, speech, translation, OCR and text-to-speech for all 22 scheduled languages, AI4Bharat helped dismantle the argument that Indian languages lack scale or relevance.

If AI4Bharat helped prove that Indian-language AI could be built at scale, Bhashini is about ensuring it can be used at scale. Conceived as a national language technology platform, Bhashini functions as the language layer of India's digital public infrastructure (DPI), providing shared APIs, datasets and services for speech recognition, translation, text-to-speech and OCR across Indian languages. Rather than a single application, it acts as an enabling backbone, allowing govern-



TAKEAWAYS

- **AI4Bharat proved that Indian languages can be treated as first-class citizens in modern AI systems, built and deployed at scale**
- **Bhashini operates as the language layer of India's digital public infrastructure, offering shared APIs and services for speech recognition, translation etc.**
- **BharatGen and Sarvam AI aim to build large, multilingual and multimodal foundation models rooted in Indian data, scripts and use cases**

Illustration by **NILANJAN DAS / AI**